Report II · SF2930 Regression Analysis

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Introduction

The travel insurance that is normally a part of your home insurance does not cover travel done as part of your occupation. Therefore, companies that require some travel buy separate travel insurance for their employees. The insurance covers a range of different situations such as lost luggage, delays, medical costs, etc. In this lab we will focus on the potential compensation that is paid out in case of cancellations.

Travel insurance has been of high interest in recent years due to the aftermath of the pandemic changing travel behaviour. If has asked you to help them review their price models and create a price model that can price their customers on the form:

$$price = \gamma_0 \prod_{k=1}^{M} \gamma_{k,i} \tag{1}$$

where γ_0 is the base premium and $\gamma_{k,i}$, $k = 1, \ldots, M$, are the risk factors corresponding to variable number k and variable group number i. $\gamma_{k,i}$ takes different values depending on that company's characteristics. For example, let $k = 1$ be the number of persons insured, which for one company is 27. Then, according to the table below, $\gamma_1 = 10$.

Normally, assigning the same factor regardless of whether the company has 11 or 29 employees makes for a very poor price formula, but for the sake of this example we are keeping it simple. In your hand-ins you typically want to fit a continuous curve that matches the output factors you get from the regression.

Number of Persons group i	Risk factor $\gamma_{1,i}$
1: # <= 1	0.7
2: # <= 5	2.2
3: # <= 30	10
4: # = 100	25
5: # >= 100	35

Material

1. Dataset

The file `GLM_KTH_Data_Train.csv` contains information on all companies with a Business Travel insurance at If P&C during 2018–2023, including claims history. The file has one row per company and risk year, as shown in the table below.

RiskYear	NumberOfPersons	FinancialRating	...	ActivityCode	Duration	NumberOfClaims	ClaimCost
2017	9	AA	...	G	0.63	1	67 099
2018	9	A	...	Missing	0.59	1	25 850
⋮	⋮	⋮	⋱	⋮	⋮	⋮	⋮

Here, *RiskYear* is the year of the insurance period, *NumberOfPersons* denotes the number of persons covered by the insurance, *FinancialRating* is a measure of the company's economic capacity and *ActivityCode* is the activity code registered on the company, which defines what segment the company is active in. For each company, there is also information regarding *Duration*. This is the share of the risk year during which the company was insured. For example, if a company only has a one-year insurance with us, from 2018-07-01 to 2019-06-30, it will be represented by two rows in the data: one with Risk year $= 2018$ and one with Risk year $= 2019$, both with Duration $= 0.5$. Finally, the number of claims and the claim cost corresponding to the insurance period are denoted by *NumberOfClaims* and *ClaimCost*.

See Appendix A for a thorough description of the contents of the dataset.

1 Project Goals

In this project, we will try to help the company If by developing a pricing model for business travel insurance. If has data on the companies that took out insurance during previous years. The main parameters are the following:

- `RiskYear`: The year of the insurance period.
- `NumberOfPersons`: The number of persons covered by the insurance.
- `FinancialRating`: A measure of the company's economic capacity.
- `ActivityCode`: The industry sector in which the company operates.
- `Duration`: The fraction of the year the insurance was active.
- `NumberOfClaims`: The number of claims filed during the insurance period.
- `ClaimCost`: The total cost of claims for the insurance period.

The main objective is to create a multiplicative Generalized Linear Model (GLM) to predict both the frequency of claims and the severity of claims. Then, the final model will be used to determine insurance premiums based on various risk factors, such as the number of insured persons, the company's financial rating, and the industry sector.

The GLM analysis will involve grouping continuous variables into risk-homogeneous categories. This grouping is essential to ensure stable and meaningful results from the GLM. The model will be evaluated using statistical tests.

2 Results

In order to fit the GLM, we had to make homogeneous groups that might be relevant. Therefore, we tried to include as many interesting groups as possible.

With the data we had, we built four relevant GLMs in order to determine which was the most accurate. First, we categorized the data into groups with homogeneous characteristics, ensuring that each category contained at least one recorded claim. The main coefficients used to evaluate the models are displayed in Table 1.

We decided to choose the model 'FinancialGroup' combined with 'AgePCompany', since the GLM coefficients were the most reasonable (for both severity and frequency): a high pseudo R^2, which shows that the model fits well, but also a high log-likelihood and a small deviance. The groups that we tried to make as homogeneous as possible are presented in Figure 1:

We notice that we had to put the 'C' financial rating in the 'Other' group because we could not create a 'Bad' financial group due to the lack of data.

We could then compute the different frequency and severity factors for each group, and consequently the final factor as the product of the severity factor and the frequency factor of the model.

Model	Log-likelihood	Deviance	Pearson χ^2	Pseudo R^2
GLM Poisson				
AgePCompany, Financial Group	−43.763	25.748	23.3	1
NumberOfPersons, Financial Group	−61.674	50.8	60.4	1
NumberOfPersons, ActivityGroup	−34.899	11.5	11	1
AgePCompany, ActivityGroup	−46.9	36.1	39.3	1
GLM Gamma				
AgePCompany, Financial Group	−149.77	0.35	0.35	0.9723
NumberOfPersons, Financial Group	−193.68	1.842	1.92	0.60
NumberOfPersons, ActivityGroup	−158.29	4.02	3.62	0.39
AgePCompany, ActivityGroup	−161.21	4.6	4.32	0.17

Table 1: Different models and their coefficients

	FinancialGroup	AgePCompany	Duration	NumberOfClaims	ClaimCost	AvgClaimCost
0	Middle	01. <4	7724.7364	26	482191.00	18545.807692
1	Middle	02. 5-14	8363.8983	29	421957.00	14550.241379
2	Middle	02. 15-30	5184.3319	21	339208.00	16152.761905
3	Middle	04. >31	2056.4901	8	72951.00	9118.875000
4	Other	01. <4	8856.3705	16	214840.00	13427.500000
5	Other	02. 5-14	3409.2545	7	45123.00	6446.142857
6	Other	02. 15-30	3741.9431	20	131658.00	6582.900000
7	Other	04. >31	2306.9245	30	185005.13	6166.837667
8	Solid	01. <4	4642.0762	16	425462.16	26591.385000
9	Solid	02. 5-14	11584.6723	73	1789808.38	24517.923014
10	Solid	02. 15-30	9912.5639	98	1637993.93	16714.223776
11	Solid	04. >31	5724.6161	165	1368295.72	8292.701333

Figure 1: Groups chosen

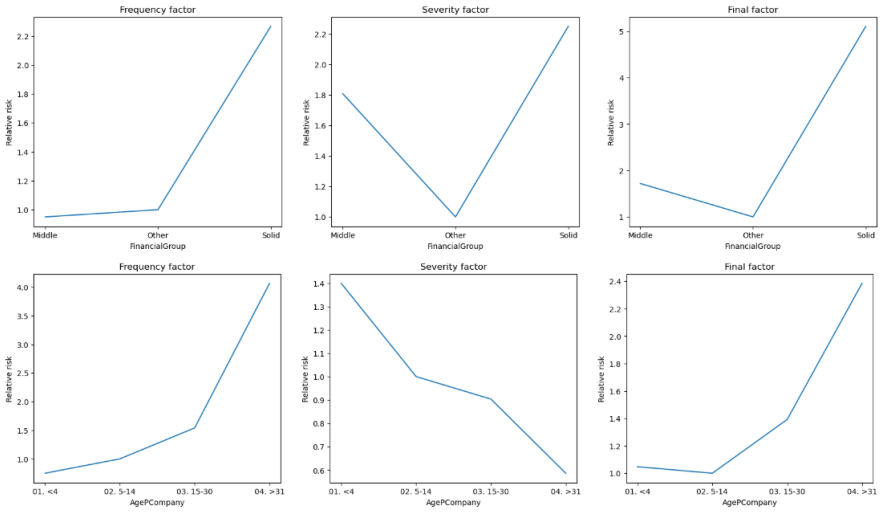


Figure 2: Factors for each group

Then, we computed the risk ratio to check whether our pricing model correctly estimates the absolute risk level between different groups.

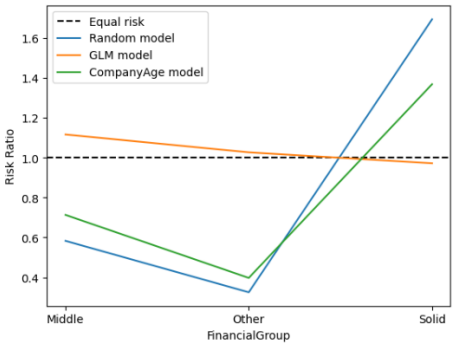


Figure 3: Risk Ratio for FinancialGroup

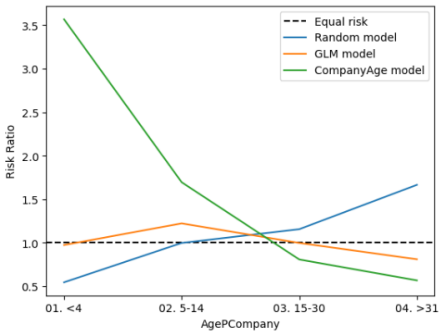


Figure 4: Risk Ratio for AgePCompany

We can see that the risk ratio for the GLM model (which is the combination of the CompanyAge and FinancialRating groups) is close to 1 for all groups, which is very satisfactory: our model accurately estimates claim costs compared to actual values.

We then wanted to determine the base level γ_0. To do so, we first try to predict the total claim cost for 2023. We assume that the insured customers will extend their contracts in 2023. Consequently,

$$ClaimCost_{2023} = \sum \frac{ClaimCost_{2022}}{Duration_{2022}} \tag{1}$$

We obtain a total claim cost for 2023 of 4193570.2. By setting a target ratio of 0.9, we obtain the companies' premium with:

$$Premium_{2023} = \frac{ClaimCost_{2023}}{0.9} \tag{2}$$

We obtain a value of 4659522.5. We then determined the risk factors for each group and insurance:

Table 2: Values of risk factors $\gamma_{k,i}$

$k \setminus i$	1	2	3	4
1	5.09852906	1.71920334	1.0	–
2	2.3845079	1.39219189	1.0	1.04737152

We notice a huge risk factor for companies that have a solid financial rating. We can see from the data that these companies have higher claim costs than the others.

We will now compute the total risk factor, which is the sum of the products of the risk factors for each contract.

For each insurance contract j, we define:

- $\gamma_{1,i}$: Financial Group risk factor (where i represents rating categories)
- $\gamma_{2,k}$: Company age risk factor

For each contract j with specific characteristics, the individual risk factor is calculated as:

$$\gamma_j = \gamma_{1,i(j)} \times \gamma_{2,k(j)}$$

where:

- $i(j)$ is the financial category for contract j
- $k(j)$ is the age category for contract j

The total risk factor γ is the sum of all individual contract risks:

$$\gamma = \sum_{j=1}^{contracts} \gamma_j$$

We determined how many companies have taken each of the following contracts:

Table 3: Risk factor calculation by group

Financial Group	**AgePCompany**	**Contracts**	**Result**
Middle	01. <4	16572	67906.25
Middle	02. 5–14	17492	41862.19
Middle	03. 15–30	10456	17975.19
Middle	04. >31	4071	7329.70
Solid	01. <4	9734	118296.88
Solid	02. 5–14	23987	170299.99
Solid	03. 15–30	19725	100568.99
Solid	04. >31	10813	57737.02
Other	01. <4	18587	44318.83
Other	02. 5–14	7326	10202.80
Other	03. 15–30	6782	6782.00
Other	04. >31	3941	4127.69
		Total	**647407.53**

Finally, the computation gives us: $\gamma = 647407.53$. We obtain the base level factor as:

$$\gamma_0 = \frac{ClaimCost_{2023}}{\gamma} = 6.48 \tag{3}$$

A The data

Variable	Description	Values
`ActivityCode`	A column describing what the company does	A – Agricultural activities B – Mining C – Manufacturing D – Production and distribution of electricity/power/heat E – Recycling F – Construction G – Wholesale & Retail stores H – Transport & Logistics I – Restaurant & hotels J – Consultants K – Finance, Insurance, etc. L – Property Owners M – Lawyers, accountants N – Leasing O – Public entities P – Education Q – Care R – Culture S – Service companies (hairdressers, repairs, etc.) T – Service companies (hairdressers, repairs, etc.) U – Mining X – Missing
`ClaimCost`	Total claim cost	
`CompanyAge`	For how long the company has been registered	
`DangerousAreas`	Describes whether travel to different dangerous areas is covered by the insurance	Excluded/Not excluded
`Duration`	The time in years that an insurance has been active during the year	
`FinancialRating`	Describes a company's credit risk. Typically geared towards the banks' risk when granting loans, but it gives some useful information about how well the company is run.	**[C − AAA]** – C is worst, AAA is best **[AN]** – Newly started company, no information **[IR]** – Has not reported the necessary documentation
`NumberOfPersons`	Number of people the insurance covers	
`NumberOfClaims`	Number of registered claims	
`RiskYear`	Which year the insurance is active	2018–2023
`TravellingArea`	Which area of travel the insurance covers	Whole world Europe Nordic countries Sweden